Estimating Sea Clutter False Alarm Probabilities with Integral Solvers

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1 Introduction

Marine radar systems transmit electromagnetic waves that bounce off objects in the ocean and the waves themselves. The echoed waves from the sea are background noise, called sea clutter, and we aim to detect objects that differ significantly from this noise. Rarely, this sea clutter can cause a spike in the radar as strong as that which a real object would. This project aims to compute the distribution of the scattered wave, measured at the radar tower, for realistic sea waves. We then want to look at the chance of the random ocean exceeding a given cutoff (i.e. a false alarm).

2 Problem formulation

We investigate the monochromatic case where we specify the wavelength λ / wavenumber k. We will use $\lambda = 10\,\text{cm}$ for our main runs.

Ellen: X-band is 3cm, paper uses either 23cm or 10cm. We should pick one and then reword introduction to be consistent

The domain we are solving on is spatially 2D (vertical z and horizontal x), and the magnetic or electric field of the incident wave must be assigned to the axis perpendicular to our plane (y), so it will be represented by a scalar. For now, we assume the ocean is a perfect electrical conductor, and we begin with the electric field case.

Let $\Gamma := \{(x, f(x))\}$ represent the surface of the sea. We wish to observe the scattering effects when an incoming field hits a particular patch of the ocean, therefore, we take f to have support in $[-1, 1]$, and call this part of the surface Γ_0 (Figure 1). We assume $f \geq 0$ so that we can apply standard techniques for the Helmholtz half-space problem, which limits realism in favor of numerical feasibility. The impose that the total field satisfies Helmholtz with Dirichlet boundary conditions:

$$(\Delta + k^2)u^{\text{tot}} = 0 \qquad \text{in } \Omega := \{(x, z) : z > f(x)\} \tag{1}$$

$$u^{\text{tot}} = 0 \qquad \text{on } \Gamma. \tag{2}$$

Since we are only interested in capturing the scattering off of Γ_0, we choose the incoming field to capture the total field off of a flat surface ($f \equiv 0$). This is given by

$$u^{\text{inc}}(\mathbf{x}) = e^{ik\mathbf{d}\cdot\mathbf{x}} - e^{ik\mathbf{d}'\cdot\mathbf{x}} \tag{3}$$

where $\mathbf{d} = (\sin\theta, -\cos\theta)$ and $\mathbf{d}' = (\sin\theta, \cos\theta)$. The scattered field then solves

$$(\Delta + k^2)u = 0 \qquad \text{in } \Omega \tag{4}$$

$$u = -u^{\text{inc}} \qquad \text{on } \Gamma_0 \tag{5}$$

$$u = 0 \qquad \text{on } \Gamma \backslash \Gamma_0 \tag{6}$$

$$\sqrt{r}\,(\partial_r u - iku) \to 0 \qquad \text{as } r \to \infty, \tag{7}$$

where we have imposed the Sommerfeld radiation condition to ensure uniqueness.

Remark 1 *There is an alternative incoming field formulation that we may choose to use instead, although we are not as sure how this would work. Following [?], we take a modified Gaussian taper as our incoming field u^{inc}, and the scattered field u is what remains:*

$$(\Delta + k^2)u = 0 \;\; \textit{in } \Omega := \{(x, z) : z > f(x)\}, \qquad u = -u^{inc} \;\; \textit{on } \Gamma, \qquad \sqrt{r}(\partial_r u - iku) \to 0 \;\; \textit{as } r \to \infty. \tag{8}$$

The taper does not exactly solve Helmholtz, and the reference reports that it fails near grazing ($\theta \to 90°$), so we take θ between $80°$ and $85°$.

3 Numerical Method and Validation

Define

$$\Phi(\mathbf{x},\mathbf{y}) = \frac{i}{4}H_0^{(1)}(k|\mathbf{x}-\mathbf{y}|), \qquad G(\mathbf{x},\mathbf{y}) = \Phi(\mathbf{x},\mathbf{y}) - \Phi(\mathbf{x},\mathbf{y}'),$$

where $\mathbf{y}' = (y_1, -y_2)$, so $G = 0$ on the flat part of Γ. Then

$$u(\mathbf{x}) = \int_{\Gamma_0} \partial_{n_\mathbf{y}} G(\mathbf{x},\mathbf{y})\varphi(\mathbf{y})ds(\mathbf{y}),$$

with n pointing into the air, satisfies the Helmholtz equation and Sommerfeld condition. Taking the limit as $\mathbf{x}$ approaches Γ_0 yields the second-kind integral equation

$$\frac{1}{2}\varphi(\mathbf{x}) + \int_{\Gamma_0} \frac{\partial G(\mathbf{x},\mathbf{y})}{\partial n_\mathbf{y}}\varphi(\mathbf{y})ds(\mathbf{y}) = -u^{\text{inc}}(\mathbf{x}) \quad \text{for } \mathbf{x} \in \Gamma_0. \tag{9}$$

The kernel $\partial\Phi(\mathbf{x},\mathbf{y}')/\partial n_\mathbf{y}$ becomes nearly singular at the endpoints of Γ_0, which we plan to treat by subtracting off the fundamental Laplace solution. The resulting integral equation can be solved with standard quadrature methods.

We plan to start with a dense solver before implementing any fast methods, and validate it first via the following methods:

1. We place a point source $\mathbf{z}$ between Γ_0 and $z = 0$. Then $u(\mathbf{x}) := G(\mathbf{x},\mathbf{z})$ automatically solves Helmholtz, the boundary condition $u(\mathbf{x}) = 0$ for $x \in \Gamma\backslash\Gamma_0$, and the Sommerfeld radiation condition. Let $g(\mathbf{x}) := u(\mathbf{x})$ for $\mathbf{x} \in \Gamma_0$. Use this g (replacing $-u^{\text{inc}}$) as the RHS of 9 and run our solver to see if we recover $u(\mathbf{x})$ for $\mathbf{x} \in \Omega$.
2. Let Γ_0 be flat ($f \equiv 0$). Then since u^{inc} already gives the total field in this case, our solver should obtain $u \equiv 0$.
3. For any $f \neq 0$, swapping the radar and receiver directions should give the same answer, and refining must reduce the error at the rate we expect from quadrature.
4. We will know the full solver is correct when it reproduces the scattering strength curve in Figure 2, which averages over random rough surfaces.

4 Ocean surfaces and False Alarm Probability

There are a variety of models we can use for the ocean surface that vary in complexity. We will begin with the simplest cases: a flat ocean, a sinusoid, and a Gaussian bump, with the last two shifted so they fit in the patch. From here, we can expand to Fourier series randomly sampled from a wave spectrum [**?**], and we will check that the average spectrum of the samples matches the target.

Far from the patch, the scattered wave looks like $A(\theta)\, e^{ikr}/\sqrt{r}$, and $A(\theta)$ is a cheap integral of φ. We assume the radar tower is far enough away that the wave measured there is given by $A(\theta)$, so each sea surface gives one number, $|A(\theta)|^2$. Monte Carlo over random sea surfaces gives the distribution, and our main deliverable is the chance of exceeding each cutoff,

$$P(|A(\theta)|^2 > \tau).$$

If Monte Carlo is infeasibly slow, we will make the solver fast using the fast multipole method.

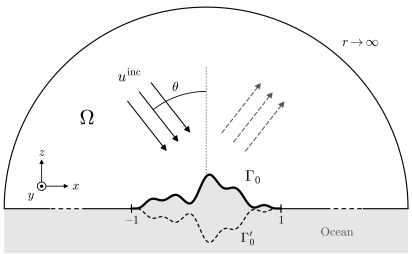


Figure 1: The computational domain, with the mirror image of Γ_0, based on Figure 2-2 from [1]

References

[1] J. V. Toporkov, *Study of Electromagnetic Scattering from Randomly Rough Ocean-Like Surfaces Using Integral-Equation-Based Numerical Technique*. PhD thesis, Virginia Polytechnic Institute and State University, Blacksburg, VA, 1998.

[2] E. I. Thorsos, "The validity of the Kirchhoff approximation for rough surface scattering using a Gaussian roughness spectrum," *The Journal of the Acoustical Society of America*, vol. 83, no. 1, pp. 78–92, 1988.

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[4] C. L. Rino and H. D. Ngo, "Numerical simulation of low-grazing-angle ocean microwave backscatter and its relation to sea spikes," *IEEE Transactions on Antennas and Propagation*, vol. 46, no. 1, pp. 133–141, 1998.

[5] J. V. Toporkov and M. A. Sletten, "Statistical properties of low-grazing range-resolved sea surface backscatter generated through two-dimensional direct numerical simulations," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 45, no. 5, pp. 1181–1197, 2007.

[6] H. W. Melief, H. Greidanus, P. van Genderen, and P. Hoogeboom, "Analysis of sea spikes in radar sea clutter data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 44, no. 4, pp. 985–993, 2006.

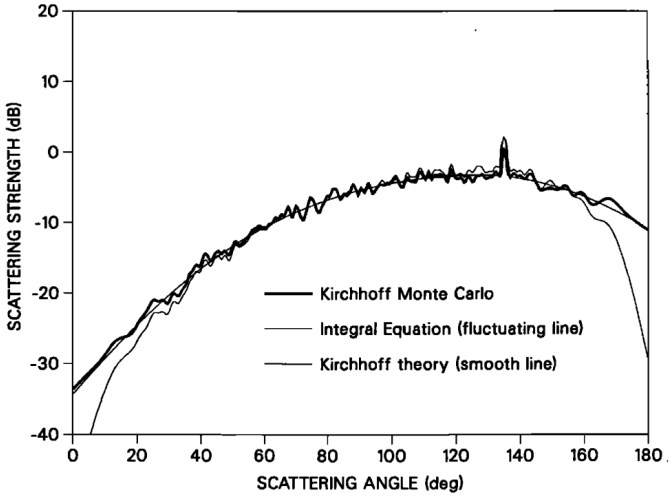


Figure 2: Figure 4 from [2], showing a comparison of scattering strength under several different solves, at varying scattering angle. Our target is the integral equation curve.